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Reg 19893

Entropía de Fuente

A) S.	p(S _i)	(I)	S _i	p(S _i)	(II)
A	4/16	A	4/16	4/20	A
B	2/16	E	4/16	4/20	B
C	1/16	B	2/16	2/20	C
D	2/16	D	2/16	2/20	D
E	4/16	C	1/16	2/20	E
F	1/16	F	1/16	2/20	F
G	1/16	G	1/16	2/20	G
H	1/16	H	1/16	2/20	H

8b por símbolo
↓
64b

B) Shannon (I)

	FA	Decimal	Binario	$\log_2(1/p(S_i))$	$(*)+1$	long	Código
A	4/16	0	0	2	3	3	000
E	4/16	0.25	0.010	2	3	3	010
B	2/16	0.5	0.1000	3	4	4	1000
D	2/16	0.625	0.1010	3	4	4	1010
C	1/16	0.75	0.11000	4	5	5	11000
F	1/16	0.8125	0.11010	4	5	5	11010
G	1/16	0.875	0.11100	4	5	5	11100
H	1/16	0.9375	0.11110	4	5	5	11110

$$\begin{aligned}
 0.25 \times 2 &= 0.5 & 0.5 \times 2 &= 1 & 0.625 \times 2 &= 1.25 & 0.75 \times 2 &= 1.5 & 0.8125 \times 2 &= 1.625 \\
 0.5 \times 2 &= 1 & & & 0.25 \times 2 &= 0.5 & 0.5 \times 2 &= 1 & 0.625 \times 2 &= 1.25 \\
 0 \times 2 &= 0 & & & 0.5 \times 2 &= 1 & & & 0.25 \times 2 &= 0.5 \\
 & & & & & & & & 0.5 \times 2 &= 1 \\
 0.875 \times 2 &= 1.75 & 0.9375 \times 2 &= 1.875 & & & & & & \\
 0.75 \times 2 &= 1.5 & 0.875 \times 2 &= 1.75 & L(S) &= 3.75 & & & & \\
 0.5 \times 2 &= 1 & 0.75 \times 2 &= 1.5 & 8 \times 3.75 &= 30b & & & & \\
 0.5 \times 2 &= 1 & 0.5 \times 2 &= 1 & & & & & &
 \end{aligned}$$

Huffman

	4/16	4/16	4/16	4/16	4/16	8/16	8/16	0	Código
A	4/16	4/16	4/16	4/16	4/16	8/16	8/16	0	10
E	4/16	4/16	4/16	4/16	4/16	8/16	8/16	1	11
B	2/16	2/16	2/16	2/16	2/16	4/16	4/16	0.1	010
D	2/16	2/16	2/16	2/16	2/16	4/16	4/16	0.1	011
C	1/16	2/16	2/16	2/16	2/16	4/16	4/16	0.1	0000
F	1/16	1/16	2/16	2/16	2/16	4/16	4/16	0.1	0001
G	1/16	1/16	2/16	2/16	2/16	4/16	4/16	0.1	0010
H	1/16	1/16	2/16	2/16	2/16	4/16	4/16	0.1	0011

$$L(H) = 2.75 \rightarrow \text{bits} \times \text{símbolo} \\
 8 \times 2.75 = 22b$$

Sr	$p(s_i)$	FA	Decimal	Binary	$\log_2(1/p(s_i))$	$\lceil \log_2(1/p(s_i)) \rceil$	long	codings
A	$4/20$	0	0	0	1.32	2	3	000
B	$4/20$	$1/20$	0.2	0.001	2.32	3	3	001
C	$2/20$	$3/20$	0.4	0.0110	3.32	4	4	0110
D	$2/20$	$10/20$	0.5	0.1000				1000
E	$2/20$	$12/20$	0.6	0.1001				1001
F	$2/20$	$14/20$	0.7	0.1011				1011
G	$2/20$	$16/20$	0.8	0.1100				1100
H	$2/20$	$18/20$	0.9	0.1110				1110

$$\begin{array}{llllll}
 0.2 \times 2 = 0.4 & 0.4 \times 2 = 0.8 & 0.5 \times 2 = 1 & 0.6 \times 2 = 1.2 & 0.7 \times 2 = 1.4 & 0.8 \times 2 = 1.6 \\
 0.4 \times 2 = 0.8 & 0.8 \times 2 = 1.6 & 0 \times 2 = 0 & 0.2 \times 2 = 0.4 & 0.4 \times 2 = 0.8 & 0.6 \times 2 = 1.2 \\
 0.8 \times 2 = 1.6 & 0.6 \times 2 = 1.2 & & 0.4 \times 2 = 0.8 & 0.8 \times 2 = 1.6 & 0.2 \times 2 = 0.4 \\
 & 0.2 \times 2 = 0.4 & & 0.8 \times 2 = 1.6 & 0.6 \times 2 = 1.2 & 0.4 \times 2 = 0.8
 \end{array}$$

$$\begin{array}{l}
 0.9 \times 2 = 1.8 \\
 0.8 \times 2 = 1.6 \\
 0.6 \times 2 = 1.2 \\
 0.2 \times 2 = 0.4
 \end{array}
 \quad
 \begin{array}{l}
 L(S) = 3.6 \\
 23.85
 \end{array}$$

Huffman

A	$4/20$	$4/20$	$4/20$	$4/20$	$8/20$	$8/20$	$12/20$	0	000
B	$4/20$	$4/20$	$4/20$	$4/20$	$4/20$	$8/20$	$8/20$	1	001
C	$2/20$	$4/20$	$4/20$	$4/20$	$4/20$	$4/20$	$4/20$	01	010
D	$2/20$	$2/20$	$4/20$	$4/20$	$4/20$	$4/20$	$4/20$	01	011
E	$2/20$	$2/20$	$2/20$	$4/20$	$4/20$	$4/20$	$4/20$	100	100
F	$2/20$	$2/20$	$2/20$	$2/20$	$4/20$	$4/20$	$4/20$	101	101
G	$2/20$	$2/20$	$2/20$	$2/20$	$2/20$	$4/20$	$4/20$	110	110
H	$2/20$	$2/20$	$2/20$	$2/20$	$2/20$	$2/20$	$4/20$	111	111

$$\begin{array}{l}
 L(H) = 3 \\
 245
 \end{array}$$

Encriptación

(8, 7, 6, 5, 4, 3, 2, 1)

A)

VPOJSPSE	TCYZJYUR	UWYHPTLE	PEIQFC LI
BLUEBLUE	BLUEBLUE	BLUEBLUE	BLUEBLUE
UEUFREYA	SREVINAN	TLEDOIRA	OTOMERRE
AYERFUEU	NANIVERS	ARIODELT	ERREMOTO

B) $H_{orig} = - \sum p(s_i) \log_2 p(s_i) = 4.01532$

s_i	$p(s_i)$	s_i	$p(s_i)$
V	1/32	G	2/32
P	4/32	W	5/32
O	1/32	V	1/32
J	2/32	J	5/32
S	2/32	C	1/32
E	3/32	A	3/32
T	2/32	S	3/32
C	2/32	F	1/32
Y	3/32	N	2/32
Z	1/32	H	2/32
U	2/32	P	1/32
R	1/32	X	1/32
W	1/32	M	3/32
H	1/32	O	1/32

$$\begin{aligned}
 2+7 &= 9 \\
 7 &= 8-1 \\
 11+11 &= 22 \\
 14 &= 25-11 \\
 11+20 &= 31 \quad 26 \\
 &\quad 05 \quad 1 \\
 H &= I - B \\
 D+U &= F - 26 \\
 L &= F - U \\
 S &+ 26
 \end{aligned}$$

C) Considero más seguro la encriptación de Vigenere, ya que rompe bastante con las prob de ocurrencias de los símbolos por lo que es más difícil descifrarlo.

D) Lo conveniente sería primero comprimir y luego encriptar, ya que los algoritmos vistos de compresión se basan bastante en encontrar la frecuencia de aparición de los símbolos, por lo que el proceso podría no ser muy conveniente luego de encriptar, ya que rompe con estas frecuencias

Original
A 3/32
Y 1/32
E 6/32
R 5/32
T 1/32
U 2/32
L 1/32
N 2/32
I 2/32
V 1/32
S 1/32
O 3/32
D 1/32
T 1/32
M 1/32

$$H_{original} = 3.60534$$

$$H_{ave} = H_{original}$$



Nota sobre esta hoja: lo que se ve borroso es la fórmula de la entropía.

Canal de comunicación

$$\begin{array}{cc} & b_1 & b_2 \\ a_1 & 0.8 & 0.2 \end{array}$$

$$a_2 \quad 0.2 \quad 0.8$$

$$\begin{aligned} p(a_1) &= 0.7 \\ p(a_2) &= 0.3 \\ p(b_1) &= 0.62 \\ p(b_2) &= 0.38 \end{aligned}$$

$$I(A, B) = H(A) - H(A/B) = H(B) - H(B/A) = I(B, A)$$

$$H(B/A) = \sum_j p(b_j) * \sum_i p(b_j/a_i) * \log_2(1/p(b_j/a_i))$$

$$p(b_1) = \sum_i p(a_i) * \sum_j p(b_j/a_i)$$

$$\begin{aligned} p(b_1) &= p(a_1) * p(b_1/a_1) + p(a_2) * p(b_1/a_2) \\ &= 0.7 * 0.8 + 0.3 * 0.2 \\ &= 0.62 \end{aligned}$$

$$\begin{aligned} p(b_2) &= 0.7 * 0.2 + 0.3 * 0.8 \\ &= 0.38 \end{aligned}$$

$$\begin{aligned} H(B/A) &= p(b_1) * p(b_1/a_1) * \log_2(1/p(b_1/a_1)) + p(b_1) * p(b_1/a_2) * \log_2(1/p(b_1/a_2)) + \\ &\quad p(b_2) * p(b_2/a_1) * \log_2(1/p(b_2/a_1)) + p(b_2) * p(b_2/a_2) * \log_2(1/p(b_2/a_2)) \\ &= 0.62 * 0.8 * \log_2(1/0.8) + 0.62 * 0.2 * \log_2(1/0.2) + 0.38 * 0.2 * \log_2(1/0.2) + \\ &\quad 0.38 * 0.8 * \log_2(1/0.8) \\ &= 0.4476 + 0.2743 \\ &= 0.7219 \end{aligned}$$

$$\begin{aligned} H(B) &= \sum_j p(b_j) * \log_2(1/p(b_j)) \\ &= p(b_1) * \log_2(1/p(b_1)) + p(b_2) * \log_2(1/p(b_2)) \\ &= 0.62 * \log_2(1/0.62) + 0.38 * \log_2(1/0.38) \\ &= 0.958 \end{aligned}$$

$$H(B) - H(B/A) = 0.958 - 0.7219 = 0.2361$$

$$I(A, B) = 0.2361$$

Cap de canal $\rightarrow p(a_1) = p(a_2) = 0.5 \quad p(b_1) = 0.5 = p(b_2)$

$$H(B) = 1$$

$$H(B/A) = 0.7219$$

$\therefore$ La capacidad de canal es $I(A, B) = 0.7219$, para $p(a_1) = p(a_2) = 0.5$

B)

	b_1	b_2	b_3	
a_1	0.3333	0.3333	0.3333	$p(a_1) = p(a_2) = p(a_3) = 0.3333$
a_2	0.3333	0.3333	0.3333	
a_3	0.3333	0.3333	0.3333	

$$p(b_1) = 0.3333 * 1 = 0.3333 = p(b_2) = p(b_3)$$

$$\begin{aligned} H(B) &= \sum_j p(b_j) * \log_2(1/p(b_j)) \\ &= (0.3333 * \log_2(1/0.3333)) * 3 \\ &= 1.5849 \end{aligned}$$

$$\begin{aligned} H(B/A) &= \sum_j p(b_j) * \sum_i p(b_i/a_i) * \log_2(1/p(b_j/a_i)) \\ &= 0.3333 * 0.3333 * \log_2(1/0.3333) \\ &= 0.5283 \end{aligned}$$

$$\begin{aligned} I(A, B) &= H(B) - H(B/A) \\ &= 1.5849 - 0.5283 \\ &= 1.0566 \leftarrow \text{capacidad de canal} \end{aligned}$$